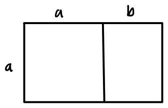
Fibonacci

mandag den 7. oktober 2024 13.26

φ fullfils these proposins:

1. 

$\frac{a+b}{a} = \frac{a}{b} = \varphi$

2. $\varphi^2 = \left(\frac{1\ \sqrt5}{2}\right)^2 = \frac{1+2\sqrt5+5}{2} = \frac{3+\sqrt5}{2} = 1+\varphi$

$n=3.$ $f_3 = 2$

$\varphi^{3-2} = \varphi \approx 1{,}6 < f_3$

$n=4$ $f_4 = 3$

$\varphi^{4-2} = \varphi^2 \approx 2{,}6 < f_4$

$n=5$ $f_5 = f_4 + f_3$

$\varphi^2 + \varphi = \varphi(\varphi+1) = \varphi^3$

Proof by (strong) induction

Base step: see above $(n=3, n=4)$

Induction hypothesis: $(k \geq 4)$

$f_{k-1} > \varphi^{k-3}$ and $f_k > \varphi^{k-2}$

Induction step: $(k \geq 4)$

$f_{k+1} = f_k + f_{k-1} > \varphi^{k-2} + \varphi^{k-3} = \varphi^{k-3}(\varphi+1) = \varphi^{k-1}$

$n = 2a + 5b$

n kroner $(n \geq 4)$ can be given as change using 2 and 5 kroner coins

$4 = 2 \cdot 2$ $(a=2, b=0)$

$5 = 5 \cdot 1$ $(a=0, b=1)$

$6 = 2 \cdot 3$ $(a=3, b=1)$

$7 = 2 \cdot 1 + 5 \cdot 1$ $(a=1, b=1)$

$8 = 2 \cdot 4$ $(a=4, b=0)$

$9 = 2 \cdot 2 + 5 \cdot 1$ $(a=2, b=1$

$P(n) = \exists a, b \in \mathbb{N} : n = 2a + 5b$

$P(4) \Rightarrow P(6) \Rightarrow P(8) \Rightarrow \ldots$

$P(5) \Rightarrow P(7) \Rightarrow P(9) \Rightarrow \ldots$

Proof by strong Induction

bas. $P(4) \land P(5)$

IH for $k \geq 6$, $P(k-2)$

IS for $k \geq$,

$k = k-2+2 = 2a+5b+2$

$k = 2(a+1)+5b$